

FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2022

FULL TEST – VIII

PAPER –1

TEST DATE: 14-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
 - Each subject (PCM) has 19 questions.
 - This question paper contains **Three Parts**.
 - **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
 - Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
- Section – A (01 – 04, 20 – 23, 39 – 42):** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Section – A (05 –10, 24 – 29, 43 – 48):** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Section – B (11 – 13, 30 – 32, 49 – 51):** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
- Section – C (14 – 19, 33 – 38, 52 – 57):** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

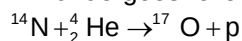
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. ^{14}N undergoes following reaction



where p is proton

Given mass of following,

$$^{14}\text{N} = 14.00307 \text{ u}$$

$$^4_2\text{He} = 4.00260 \text{ u}$$

$$1 \text{ amu} = 931 \text{ MeV} / c^2$$

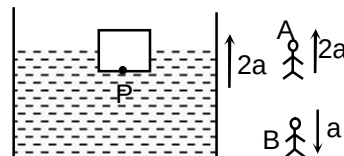
$$^{17}\text{O} = 16.999134 \text{ u}$$

$$\text{p} = 1.007834 \text{ u}$$

Choose the correct option.

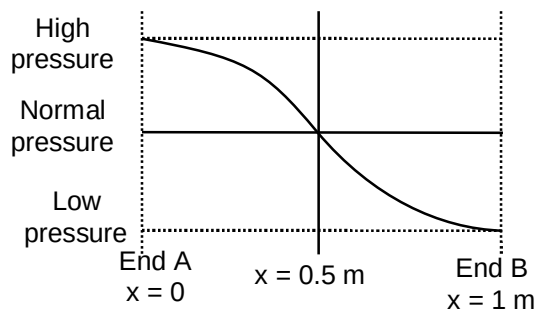
- (A) energy released in process is 1.40 MeV
- (B) energy released in process is 1.20 MeV
- (C) energy released in process is 1.10 MeV
- (D) energy released in process is 1.50 MeV

2. A block of density ρ is floating in liquid of density σ as shown. Container is accelerating upwards with acceleration $2a$. There are two observers A and B as shown in the figure. Observer A is accelerating upward with acceleration $2a$ while observer B is accelerating downwards with acceleration a . Choose the correct option.



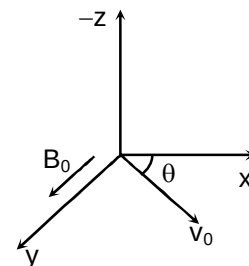
- (A) Pressure at point P will more as per observer A compare to observer B.
- (B) Buoyant force on block will be more for A compare to B.
- (C) Buoyant force on block will be same for both observer.
- (D) Both option (A) and option (B) are correct.

3. A pipe of length 1 m is sounded with some mechanism. During propagation of wave, pressure at resonance varies inside pipe at $t = 1 \text{ sec}$ as shown in the figure. Choose the correct option.



- (A) End A of pipe is open while end B is closed.
- (B) End A of pipe is closed while its end B is open
- (C) Both ends of pipe are open.
- (D) Both ends of pipe are closed.

4. A uniform magnetic field exists in region along y-axis. A charge $+q$ is projected at origin in xy plane with velocity v_0 at $t = 0$ as shown. Choose the correct option for motion of charge. ($\theta \neq 0$)

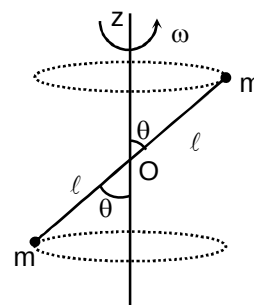


- (A) x and z coordinate of position of charge can not be zero at same time ($t > 0$)
 (B) If speed v_0 is increased, charge will take more time to have its x-coordinate to be zero for the first time after $t = 0$.
 (C) If θ is increased keeping speed v_0 constant, charge will take same time to meet y-axis after $t > 0$.
 (D) all of the above

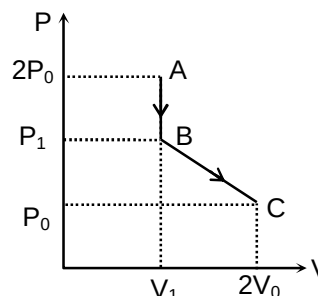
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. Two point masses each of mass m are attached to ends of a massless rod of length 2ℓ . z-axis passes through mid point O of rod and vertical. Rod is fixed at point O and rotates about z-axis with constant angular velocity ω . Which of the following statement(s) is/are incorrect?



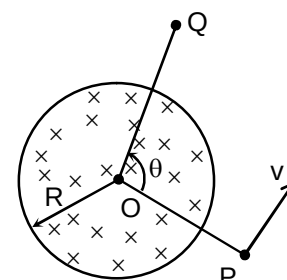
- (A) Torque on system is zero
 (B) Angular momentum of system ($\vec{L}$) and its angular velocity $\vec{\omega}$ are parallel as $\vec{L} = I\vec{\omega}$
 (C) Angular momentum of system is $2m\omega\ell^2\cos^2\theta$.
 (D) Angular momentum of system ($\vec{L}$) is related to torque acting on system as $\vec{\tau} = \frac{d\vec{L}}{dt}$ about any point (on the system or off the system)
6. A student designs a vernier calipers with least count 0.3 mm. Main scale of vernier calipers has one main division 0.1 cm. If A be the number of divisions on vernier scale and B is length of vernier scale.
- (A) $A = 10, B = 7$ mm
 (B) $A = 15, B = 10.5$ mm
 (C) $A = 15, B = 9$ mm
 (D) $A = 10, B = 10.5$ mm
7. A monoatomic gas consisting of n moles is taken from state A to state C as shown in the figure. Process $A \rightarrow B$ is isochoric which of following is/are correct.



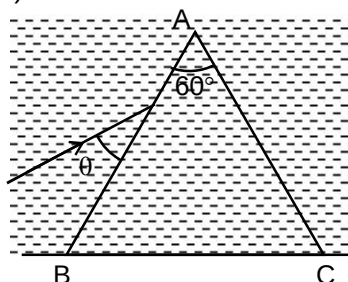
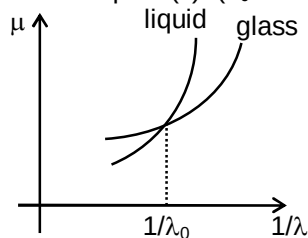
- (A) Temperature of gas at state B is minimum for $P_1 = \frac{3P_0}{2}$ and $V_1 = \frac{3V_0}{2}$
- (B) For $P_1 = \frac{3P_0}{2}$, $V_1 = \frac{3V_0}{2}$, minimum internal energy of gas will be $3P_0V_0$.
- (C) For $P_1 = \frac{7P_0}{4}$, $V_1 = \frac{3V_0}{2}$, heat absorbed/released from $A \rightarrow C$ will be more compare to heat released/absorbed from $A \rightarrow C$ when $P_1 = \frac{3P_0}{2}$, $V_1 = \frac{7V_0}{4}$
- (D) In process $B \rightarrow C$, temperature of gas first decreases and then increases if $P_1 = \frac{3P_0}{2}$, $V_1 = \frac{3V_0}{2}$

8. A long and tightly wound solenoid has 100 turns in its one meter length and current I flows through its winding increasing at rate of $\frac{5 \times 10^{-2}}{\mu_0}$ A/s. Area of cross section of mid part of solenoid is

shown in figure. Assuming a uniform magnetic field over the cross section. A charge having $q = 2 \mu\text{C}$ and $m = 1 \text{ mg}$ is projected with velocity v perpendicular to OP and in plane of cross section. Charge reaches at point Q at time t . OQ makes angle θ with OP as shown. Which of the following is incorrect? (Assume there is no magnetic field outside solenoid)

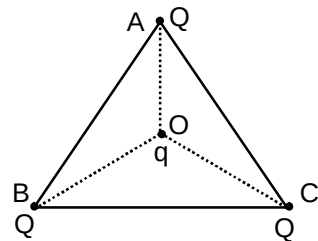


- (A) Acceleration of charge at point P is zero for v to be zero.
- (B) For $\theta = \frac{\pi}{3}$, velocity of charge at Q is 3.2 m/s for $v = 0$, $\frac{dB}{dt} = 5 \text{ T/s}$, $R = 1 \text{ m}$.
- (C) For larger value of I in solenoid charge will gain greater speed for given θ
- (D) $OP = r$, for $r > R$, speed of charge at point Q is independent of R .
9. A glass prism having angle of prism 60° is placed inside liquid of refractive index μ_ℓ . Refractive index of glass is μ_g . Variation of μ_ℓ and μ_g with wavelength λ are shown in figure. Choose the incorrect option(s). ($\lambda_0 \rightarrow$ wavelength of green light)



- (A) For $\theta = 30^\circ$, blue colour ray will go undeviated.
- (B) For $\theta = 87^\circ$, yellow colour light will deviate towards vertex A .
- (C) A white light incident on face AB , all colours will be deviated towards base BC for $\theta = 80^\circ$.
- (D) For light having $\lambda < \lambda_0$, ray will bend towards base BC .

10. Three equal charges $+Q$ are fixed at vertices of equilateral triangle of side a . Another charge $+q$ is placed at centroid of triangle. Choose the correct option.

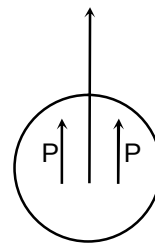


- (A) Centroid, O of triangle is position of unstable equilibrium for charge q .
 (B) If charge q is displaced slightly along AO and released charge q will undergo oscillatory motion.
 (C) If charge q is displaced along normal to plane of triangle, charge q will gain velocity $\sqrt{\frac{3\sqrt{3}Q}{2\pi\epsilon_0 m a}}$ finally, where m is mass of charge q .
 (D) Electric field is always a conservative force field.

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. When a dielectric body is placed in external electric field, it gets polarized. Surface charge density of body is given as $\sigma_b = \vec{P} \cdot \hat{n}$, where $\vec{P}$ dipole moment created per unit volume and $\hat{n}$ is radially outward unit vector at surface of the body. Volume charge density at any point of the body is $\rho_b = -\nabla \cdot \vec{P}$, where $\nabla = \left(\frac{d}{dx} \hat{i} + \frac{d}{dy} \hat{j} + \frac{d}{dz} \hat{k} \right)$. A solid spherical body gets uniformly polarized in external electric field with constant dipole moment per unit volume P as shown. Radius of sphere is R . What is electric field at distance one meter from center of sphere is $3x$ N/C.



Find x . (given $R = 2\text{m}$, $P = \frac{10^{-9}}{\pi} \text{ C/m}^2$)

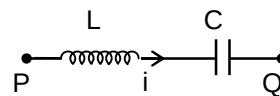
12. Two longitudinal waves, one $y_1 = A \sin(\omega t - kx)$ moving along +ve x -axis and another $y = A \sin(\omega t - ky)$ moving along +ve y -axis, superimpose. Some of the particle of medium move on circular path. Locus of these particles is $y = x \pm \left(n \pm \frac{1}{m} \right) \lambda$. Find the value of m .
13. A disc of radius R is placed in vertical plane on rough horizontal surface ($\mu_k = 0.5$). A string of length R is tied at circumference of disc while its other end is kept at level of centre of disc from ground. String is pulled such that disc is in equilibrium. If string makes angle θ with horizontal then $\sin \theta = \frac{1}{x}$, find x .

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15
Question Stem

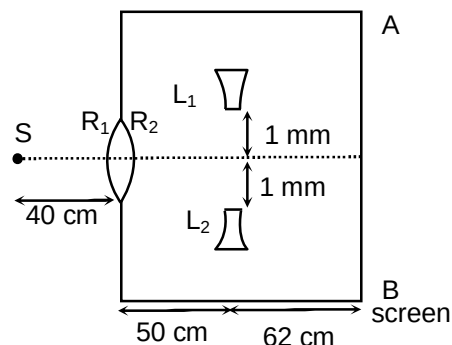
An inductor $L = 20 \text{ mH}$ and a capacitor $C = 50 \text{ mF}$ are connected in series as shown in the figure. At time $t = t_0$ current in branch PQ is 1 A and is increasing at a rate 50 A/s . Time at which potential difference across PQ is zero at $(t_0 + X) \text{ sec}$. In another case, current in branch across PQ is 1 A at $t = 0$ and is decreasing uniformly at rate 10 A/s , time at which voltage across PQ is zero, is $Y \text{ sec}$. (Take $\pi = 3$ and capacitor is initially uncharged) (potential difference across PQ changes sinusoidally with $\omega = 50 \text{ rad/s}$)



14. The value of X is
15. The value of Y is

Question Stem for Question Nos. 16 and 17
Question Stem

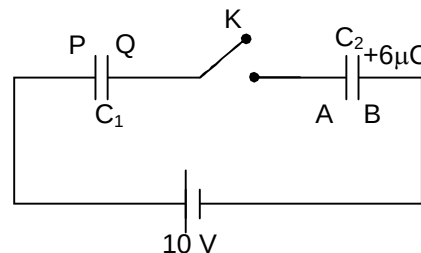
A cubical container is filled with liquid of refractive index 1.2 . A convex lens having radius of curvature $R_1 = 20 \text{ cm}$ and $R_2 = 24 \text{ cm}$ is fixed in one face of cuboid. Parts of concave lens having (concave lens is cut into two equal parts along its principal axis) having focal length 20 cm in liquid are placed as shown (refractive index of material of concave lens 1.15). An interference pattern is obtained on screen AB for source S. Distance between images formed by parts of concave lens is $X \text{ mm}$ and fringe width of pattern formed on the screen is $Y \text{ mm}$. (Consider rays passing through parts of the concave lens only, λ for source is 500 nm in liquid and refractive index of material of convex lens is 1.5)



16. The value of X is
17. The value of Y is

Question Stem for Question Nos. 18 and 19
Question Stem

Two capacitors $C_1 = 3 \mu\text{F}$ and $C_2 = 6 \mu\text{F}$ are connected as shown in the figure. When K is open C_1 is unchanged while plate B of C_2 is given charge $6 \mu\text{C}$. After key is closed, magnitude of charge on plate P of capacitor C_1 is $X \mu\text{C}$ and magnitude of charge on plate B of C_2 , is $Y \mu\text{C}$.



18. The value of X is
19. The value of Y is

Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. What is not produced in given reaction change?



- (A) H_2
 (B) NaOH
 (C) Hg
 (D) O_2

21. The compound formed by dissolving gold and platinum in aqua-regia is

- (A) $[\text{AuCl}_4]^{3-}$ and $[\text{PtCl}_6]^{2-}$
 (B) $[\text{AuCl}_4]^-$ and $[\text{PtCl}_6]^{4-}$
 (C) $[\text{AuCl}_4]^-$ and $[\text{PtCl}_6]^{2-}$
 (D) $[\text{AuCl}_4]^-$ and $[\text{PtCl}_6]^{3-}$

22. What is the emf of a cell containing two hydrogen electrodes, the negative one in contact with 10^{-6} M OH^- and the positive one in contact with a $0.1 \text{ M CH}_3\text{COONH}_4$ solution?

Given $\text{pK}_a (\text{CH}_3\text{COOH}) = 4.74$ and $\text{pK}_a (\text{NH}_4^+) = 9.26$

- (A) 0.059 volt
 (B) 0.59 volt
 (C) 0.0059 volt
 (D) 5.9 volt

23. $\text{PhCONH}_2 + \text{H} \begin{array}{c} \text{Me} \\ | \\ \text{---} \\ | \\ \text{CONH}_2 \end{array} \text{D} \xrightarrow[\Delta]{\text{Br}_2 + \text{KOH}} ?$

Which of the following statement is incorrect regarding the given reaction?

- (A) PhNH_2 and $\text{H} \begin{array}{c} \text{Me} \\ | \\ \text{---} \\ | \\ \text{NH}_2 \end{array} \text{D}$ are produced

- (B) Cross-over products PhNH_2 and $\text{H} \begin{array}{c} \text{Me} \\ | \\ \text{---} \\ | \\ \text{*NH}_2 \end{array} \text{D}$ are also produced

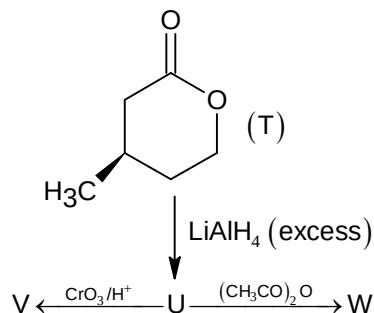
- (C) Migration of $-\text{R}$ (alkyl group) of $\text{R} - \text{CONH}_2$ takes place with retention of configuration.

- (D) PhCONHBr and $\text{H} \begin{array}{c} \text{Me} \\ | \\ \text{---} \\ | \\ \text{CONHBr} \end{array} \text{D}$ are the intermediates formed

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24. With reference to the scheme given, which of the following statement(s) about T, U, V and W is/are correct?



- (A) T is soluble in hot aqueous NaOH
 (B) U is optically active
 (C) Molecular formula of W is $\text{C}_{10}\text{H}_{18}\text{O}_4$
 (D) V gives effervescence on treatment with aqueous NaHCO_3
25. Which of the following compound(s) does not give blood red colour in Lassaigne's test ?
- (A) N#Cc1ccccc1
 (B) NC1=CC=C(C(=O)O)C=C1
 (C) NC(=O)N
 (D) NC1=CC=C(S(=O)(=O)[Na])C=C1

26. In the given question K_a values of five weak acids are given. K_a for $\text{HA} = 10^{-8}$, $\text{HB} = 10^{-6}$, $\text{HC} = 2 \times 10^{-8}$, $\text{HD} = 10^{-10}$, $\text{HE} = 10^{-7}$.

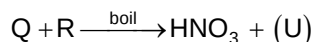
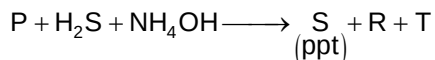
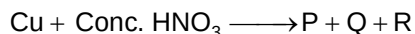
Select the correct statement(s) as per the information. Given : $\log 2 = 0.3$.

- (A) Among NaA, NaB, NaC, NaD and NaE, the NaD is most extensively hydrolysed.
 (B) Among NaA, NaB, NaC, NaD and NaE, the NaB is most extensively hydrolysed.
 (C) pH of 0.1 (M) aqueous solution of NaB is approximately 9.5.
 (D) 0.1 (M) solution of HE and 0.01 (M) solution of HB are isohydric to each other approximately.
27. The incorrect statement among the following is (I.P. is ionisation potential)
- (A) $I.P_1$ of Al < $I.P_1$ of Mg
 (B) $I.P_2$ of Mg > $I.P_2$ of Na
 (C) $I.P_1$ of Ba < $I.P_1$ of Cs
 (D) $I.P_3$ of P > $I.P_3$ of Cl

28. Which of the following oxidizes glucose into gluconic acid?

- (A) Fehling's solution
- (B) Tollen's Reagent
- (C) Nitric Acid
- (D) Bromine Water

29. Consider the following unbalanced reactions



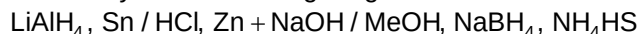
Choose the correct statement(s)

- (A) S is black in colour
- (B) Oxidation state of Cu in S is +1
- (C) R is a V-shaped molecule
- (D) Bond order of U in ground state is 2.5

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. How many of the following reagents reduce nitrobenzene in azobenzene?



31. Difference between n^{th} and $(n + 1)^{\text{th}}$ Bohr's radius of H-atom is equal to its $(n - 1)^{\text{th}}$ Bohr's radius. The value of n is

32. For isothermal expansion of an ideal gas into vacuum, among the following, how many are zero?



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

An ideal mixture of liquids A and B with 2 moles of A and 2 moles of B has a total vapour pressure of 1 atm at a certain temperature. Another mixture with 1 mole of A and 3 moles of B has vapour pressure greater than 1 atm. When 4 moles of C are added to 2nd mixture, the vapour pressure comes down to 1 atm. Vapour pressure of C in pure state $p_c^0 = 0.8$ atm. If vapour pressure of pure A = x atm and vapour pressure of pure B = y atm.

33. The value of x is _____?

34. The value of y is _____?

Question Stem for Question Nos. 35 and 36

Question Stem

If x is the total number of geometrical isomers present in the octahedral complex $[Ma_2b_2c_2]$ and y is the total number of optical isomers present in the octahedral complex $[M(AB)_3]$, where a, b, c are monodentate ligands and AB is bidentate ligand

35. The value of x is _____

36. The value of y is _____

Question Stem for Question Nos. 37 and 38

Question Stem

Potassium crystallizes in a body-centred cubic lattice, with an unit cell length $a = 5.20 \text{ \AA}$. If the distance between nearest neighbours is ' x ' and the distance between next nearest neighbour is y , then

37. The value of x is _____

38. The value of y is _____

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. The value of the expression $\frac{\sin 20^\circ (4 \cos 20^\circ + 1)}{\cos 20^\circ \cdot \cos 30^\circ}$ is
- (A) 2
(B) $\frac{1}{3}$
(C) $\frac{1}{2}$
(D) 3
40. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to
- (A) $\frac{-1}{1 + \cot^3 x} + C$
(B) $\frac{1}{3(1 + \tan^3 x)} + C$
(C) $\frac{-1}{3(1 + \tan^3 x)} + C$
(D) $\frac{1}{1 + \cot^3 x} + C$
(where C is a constant of integration)
41. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A + Bx)(x - A)^2$, then the ordered pair (A, B) is equal to
- (A) (4, 5)
(B) (-4, -5)
(C) (-4, 3)
(D) (-4, 5)
42. The length of the projection of the line segment joining the points (5, -1, 4) and (4, -1, 3) on the plane, $x + y + z = 7$ is
- (A) $\sqrt{\frac{2}{3}}$
(B) $\frac{2}{\sqrt{3}}$
(C) $\frac{2}{3}$
(D) $\frac{1}{3}$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. Let $L = \lim_{n \rightarrow \infty} \int_a^{\infty} \frac{ndx}{1+n^2x^2}$ where $a \in \mathbb{R}$, then L can be
- (A) π
 (B) $\frac{\pi}{2}$
 (C) 0
 (D) $\frac{\pi}{3}$
44. If $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sin 2x + 3y \cot x$ and $y\left(\frac{\pi}{2}\right) = 2$, then which of the following statement(s) is/are CORRECT?
- (A) $y\left(\frac{\pi}{6}\right) = 0$
 (B) $y'\left(\frac{\pi}{3}\right) = \frac{9-3\sqrt{2}}{2}$
 (C) $y(x)$ increases in $\left(\frac{\pi}{6}, \frac{\pi}{2}\right)$
 (D) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} y(x) dx = 0$
45. PQ is a double ordinate of $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to x-axis at G, then locus of G is a parabola with
- (A) vertex at $(3a, 0)$
 (B) focus at $(5a, 0)$
 (C) directrix is $x - 3a = 0$
 (D) length of latus rectum is $4a$
46. Let $I = \int_0^1 \sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} dx$ and $J = \int_0^1 \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$, then which of the following statement(s) is/are CORRECT?
- (A) $I + J = \pi$
 (B) $I - J = \pi$
 (C) $I = \frac{2+\pi}{2}$
 (D) $J = \frac{4-\pi}{2}$
47. If B is an idempotent matrix and $A = I - B$, then
- (A) $A^2 = A$
 (B) $A^2 = I$
 (C) $AB = 0$
 (D) $BA = 0$

48. If $0 < x < \frac{\pi}{2}$ and $\sin^n x + \cos^n x \geq 1$, then 'n' may belong to the interval
- (A) $[1, 2]$
 (B) $[3, 4]$
 (C) $(-\infty, 2]$
 (D) $[-1, 1]$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. Let z_r be the roots of the equation $\sum_{r=0}^{50} (z)^r = 0$, $r = \{1, 2, 3, \dots, 50\}$. If $\sum_{r=1}^{50} \frac{1}{(z_r - 1)} = -5\lambda$, then λ equals to
50. Given that the three points where the curve $y = bx^2 - 2$ intersects the x-axis and y-axis form an equilateral triangle, then $(2b - 1)$ equals
51. If in a triangle ABC $\tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$ are in Harmonic Progression, then the least value of $\cot^2 \frac{B}{2}$ is equal to

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

If $2x^3 + ax^2 + bx + 4 = 0$ (a and b are positive real numbers) has three real roots

52. The minimum value of b^3 is
53. The minimum value of $(a + b)^3$ is

Question Stem for Question Nos. 54 and 55

Question Stem

Suppose $f(x)$ and $g(x)$ are two continuous functions defined for $0 \leq x \leq 1$. Given $f(x) = \int_0^1 e^{x+t} f(t) dt$ and

$$g(x) = \int_0^1 e^{x+t} g(t) dt + x$$

54. The value of $f(1)$ is equal to
55. The value of $\left[f(0) - \frac{1}{g(0)} \right]$ is equal to (where $[.]$ denotes the greatest integer function)

Question Stem for Question Nos. 56 and 57

Question Stem

Let S_1 and S_2 be two fixed externally tangent circles with radius 2 and 3 respectively. Let S_3 be a variable circle internally tangent to both S_1 and S_2 at point A and B respectively. The tangents to S_3 at A and B meet at T. If $TA = 4$

56. The radius of circle S_3 is
57. If the area of circle circumscribing $\triangle TAB$ is $k\pi$, then k is